

1. What we are trying to solve

This project applies network science to analyze the Canadian air travel system as a directed, weighted network. Our goals are to model airport connectivity across Canada, measure centrality and clustering to identify key hubs and subnetworks, examine how network structure affects delay propagation, simulate node disruptions to evaluate network resilience, and investigate seasonal changes in network connectivity and delays.

2. Research Questions

1. Centrality and Delay Propagation: How does the structural centrality of Canadian airports correlate with their role in propagating delays across the national air travel network?
2. Network Robustness: How does the removal or disruption of highly central airports affect the overall connectivity and efficiency of the Canadian air transportation network?
3. Regional Clustering: How does the clustering of Canadian airports into regional subnetworks affect the containment or amplification of delays within the national network?
4. Temporal and Seasonal Patterns: How are Canadian airports' network connectivity and delay patterns affected by seasonal changes throughout the year?

3. Data Sources

We will collect data from the following open and reliable datasets:

- OpenFlights Database (openflights.org/data.html) - Provides airport and route data (nodes and edges).
- Transport Canada Open Data (open.canada.ca) - Provides domestic flight performance and delay statistics.
- Optional: Other available open source websites for additional data collection.

4. Network Definition

- Nodes: Canadian airports (identified by IATA code). Attributes: Location, average delay, passenger traffic.
- Edges: Direct flight routes between airports. Attributes: Flight frequency, average delay, distance.

5. Metrics

- Node-level metrics: Degree centrality (in/out), betweenness centrality, closeness centrality, eigenvector centrality
- Global metrics: network density, average path length, diameter, efficiency
- Community metrics: modularity, cluster coefficients
- Temporal metrics: seasonal centrality comparison, delay variation across time windows

6. Planned Analyses

- Build network using R
- Compute centrality and correlate with delay data
- Detect communities (regional subnetworks)
- Simulate disruptions by removing top-central airports
- Compare metrics across different seasons
- Visualize network structure and delay propagation patterns

7. Timeline

Oct 17-24 - Collect and clean data

Oct 25-Nov 2 - Build networks and compute metrics

Nov 1 - 8 - Continue building network

Nov 3 - 10 - Run analysis

Nov 11 - 18 - Create visualizations (charts, etc.)

Nov 25 - Dec 2 - Summarize findings and create video presentation

Dec 2 onward - Finalize presentation.

Each member of the group will contribute to all phases of the project, while focusing on specific areas of responsibility. **Member A** will lead data collection, cleaning, and integration, as well as constructing the initial network model. **Another member** will focus on calculating network centrality metrics and analyzing their correlation with delay data. **A third member** will conduct community detection and study regional clustering effects on delay propagation. **The final member** will be responsible for simulating disruptions by removing highly central nodes, performing seasonal analyses, and creating network visualizations. All group members will collaborate on interpreting results, preparing the final report, and contributing to the presentation.